Research Paper Study and Partial Reproduction: Recommendation Systems

Each group will be assigned a recent research paper related to Recommendation Systems. The group is expected to:

1. Read and understand the assigned research paper.
2. Identify the research problem, motivation and research gap.
3. Understand and explain the proposed recommendation technique.
4. Select one core technique/component from the paper.
5. Implement the selected technique.
6. Conduct experiments using the dataset used in the paper, wherever practically feasible.
7. Compare the implementation with an appropriate baseline.
8. Critically analyze the results, limitations and applicability of the proposed approach.

Objectives

By completing this assignment, students will be able to:

1. Analyze recent research literature in the field of recommendation systems.
2. Identify the problem, research gap, objectives and contributions of a research paper.
3. Explain the architecture and working principle of a modern recommendation technique.
4. Implement a selected component/technique from a published research paper.
5. Evaluate the implementation using appropriate datasets, metrics and baselines.
6. Critically discuss the strengths, limitations, assumptions and possible extensions of the proposed approach.

Tasks

Part A – Research Paper Understanding

A1. Problem Definition

- What recommendation problem is being addressed?
- Is it:
    - rating prediction?
    - Top-N recommendation?
    - sequential recommendation?
    - multimodal recommendation?
    - personalized recommendation?
    - cold-start recommendation?
    - explainable recommendation?

A2. Motivation

Why is the problem important?

A3. Research Gap

What limitations of existing methods does the paper attempt to address?

A4. Contributions

Identify major contributions claimed by the authors.

Part B – Methodology Analysis

1. Input

What information enters the model?

For example:

User ID + Item ID + Review Text + Image

2. Processing

Show the important steps:

Input → Feature Extraction → Representation → Interaction → Prediction

3. Output

What does the model finally predict?

For example:

Rating / Top-K items / Probability / Ranking score

Part C – Select One Technique for Implementation

Select and implement one technically meaningful component of the assigned paper.

Minimum requirements

The selected component should:

- be clearly identifiable in the paper;
- contribute to the recommendation process;
- be implemented in the paper rather than merely calling a complete existing implementation;
- be experimentally evaluated.

Require a Baseline

The assignment can state:

At least one baseline is mandatory.

Dataset Requirement

- Use the same dataset used in the assigned paper**.**
- If the exact dataset is publicly available and computationally feasible.
- **If not feasible**
- an officially available subset of the same dataset
- or
- an alternative dataset approved by me with similar characteristics.

Ablation/Analysis Component
Add a small ablation study based on the paper

Critical Evaluation

- Strengths
    - What does the proposed method do well?
- Limitations
    - What are its limitations?
- Computational requirements

    - Training time
    - GPU requirement
    - Number of parameters
    - Memory requirements, if relevant

Data requirements

- Does the method require:
    - reviews?
    - images?
    - user history?
    - side information?

Generalizability

Would the approach work in another recommendation scenario?

Reproducibility

Could another researcher reproduce the reported results from the information/code provided?

Possible improvement

Suggest one meaningful improvement**.**

10-Mark Rubric

Component	Marks
1. Paper understanding & research problem	1.5
2. Methodology/algorithm analysis	1.5
3. Implementation of selected technique	2.5
4. Experimental evaluation & comparison	2.0
5. Critical analysis & proposed improvement	1.5
6. Report quality, presentation & individual contribution	1.0
Total	10

Submission Guidelines

A. Part A and Part B
B. Source Code

Submit:

- Python code
- Jupyter notebook / .py files
- requirements.txt
- README

C. Results
D. Presentation and Viva